Assignment Submission

Specification-Driven Development and AI-Assisted Engineering

Student Submission Template

Student Details

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Video Link	AI-SDLC-Assignment/Final Demo.mp4 at main · AjayMurala-Bits/AI-SDLC-Assignment · GitHub https://drive.google.com/drive/u/0/folders/1SiCX1gHrjwDbX8V96g2Ln7NuanBHprdw

Problem Statements

Part 1A: Problem Statement

Provide a concise description of the problem addressed in Part 1A, including the intended users, key requirements, and expected outcome.

Response: Part 1A addresses the simple movie booking CLI application. Requirement is to develop a multi seat booking system (book all requested or nothing – no partial booking is allowed) and does not allow double booking. Intended Users are personas who wanted to view movie shows, check availability of seats, book and cancel tickets.

Part 1B: Problem Statement

Provide a concise description of the problem addressed in Part 1B, including the non-deterministic behavior, expected output, and evaluation approach.

Response: Part 1B addresses the problem of reading movie reviews by developing an AI Movie Review Summarizer. The non-deterministic behaviour is producing different worded summaries. Expected output is summary and positive and negative sentiment analysis of reviews. Evalauted based on 16 use cases with schema validation, deterministic checks. All unit tests must pass.

Assessed Reflection Questions

Answer all five questions. Each question carries 1 mark. Responses should be concise, evidence-based, and refer to your own specification artifacts, harness results, evaluation cases, or repair history. Generic descriptions of SDD or AI tools without evidence from your implementation will receive limited credit.

1. Specification influence (1 mark)

Identify one concrete decision in your implementation that was directly shaped by a Spec Kit artifact (constitution, specification, clarification, plan, or tasks). Explain the traceability from the artifact to the implemented decision.

Response: **the spe.md mentions to prevent double booking and that was the clear specification or constraint defined in the spec. Harness_log.md shows the tracebility of booking A1 and A3 seats. When bokking A1 again, it was rejected. The setas were cancelled and then shown as available when viewing the seats.**

2. Harness value in Part 1A (1 mark)

Describe one issue detected by the deterministic harness that would have been missed or less reliably detected by simply asking the coding agent to review its own output. State the relevant harness gate/evidence and how the issue was resolved.

Response: **Ruff lint check was included in the harness. Initial check or gate was failed and it was evident from the evidence log. After fixing the proper import in python script, the ruff lint check passed**

3. Evaluating non-determinism in Part 1B (1 mark)

Choose one evaluation case where exact string matching would be inappropriate. Explain which schema constraint, invariant/property, and semantic/quality criterion you used instead, and why these checks were suitable.

Response: **Exact string matching is unsuitable as the model can produce different wording for given same input**

4. Feedback and self-healing (1 mark)

Using one actual repair cycle from either part, explain what failure evidence was fed back to the coding agent, what changed in the next attempt, and whether the repair solved the root cause without weakening the specification or harness.

Response: **Part 1 A Ruff lint issue. Initial run showed failure. After fixing and re run, the gate was passed.**

5. Human accountability (1 mark)

Identify one limitation or residual risk that remained even after the harness reported PASS. Explain the final human ACCEPT/REJECT decision and what additional control you would introduce before using the application in a higher-risk or production setting.

Response: Movie booking CLI App is a single user and it does not have production ready code to support concurrent multiple uses. I accepted only for a single user. Adding distributed atomica transactions would help in concurrent bookings.